

vision-language-model-benchmark: cross-provider VLM evaluation

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2026-06-06

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1 Abstract

We present `vision-language-model-benchmark`, a cross-provider VLM evaluation harness that runs Claude / GPT-4V / Gemini / Qwen-VL / LLaVA across DocVQA, ChartQA, and MMMU-style tasks. It ships per- model accuracy / cost / latency profiles so the suite runs in CI without

API keys (a MockVLMProvider per model), and a real provider drop-in is a one-line change. We report a 450-call mock bench across 5 providers × 3 tasks × 30 items: GPT-4o tops at 0.644 accuracy (\$1.08 total), Claude-3.5-Sonnet close behind at 0.633 for \$0.77 (the cost-efficient pick), Gemini-1.5-Pro at 0.522 / \$0.50; open-source mock Qwen-VL-7B and LLaVA-1.5-13B at 0.42 and 0.28 (free). All five chart types populate, including a reliability diagram that shows the expected confidence-vs-accuracy curves.

2 1. Background

Vision-Language Model selection in production is bottlenecked by the same problem as text-only LLM selection: there are several frontier models from different providers with materially different price points, and the public leaderboards rarely report cost. A practical eval suite must report:

- per-(provider, task) accuracy
- per-question-type accuracy (counting vs reasoning vs OCR)
- per-difficulty accuracy (easy vs hard items)
- cost per item
- latency p50/p99
- confidence calibration (do the model’s reported confidences correlate with accuracy?)

This project ships all six.

3 2. Related Work

- **MMMU** (Yue et al., 2024): the major multi-discipline VLM benchmark.
- **DocVQA** (Mathew et al., 2021): document visual QA.
- **ChartQA** (Masry et al., 2022): chart understanding.
- **PaliGemma** (Beyer et al., 2024): one of the major open-weight VLMs.
- **Qwen-VL** (Bai et al., 2023): the Alibaba open-weight VLM.
- **LLaVA** (Liu et al., 2023): the canonical open VLM.

4 3. Method

4.1 3.1 Mock provider profiles

Each mock model has a `_Profile(base_acc, difficulty_penalty, base_lat_ms, cost_per_call_usd)`. The mock returns “correct” with probability $\max(0.05, \text{base_acc} - (\text{difficulty} - 1) * \text{difficulty_penalty})$, where difficulty is encoded in the synthetic image path. Confidence is a noisy version of correctness probability, so the reliability-diagram axis is meaningful.

model	base_acc	difficulty_penalty	base_lat	cost/call
claude-3.5-sonnet	0.78	0.06	600ms	\$0.0085
gpt-4o	0.81	0.05	750ms	\$0.0120
gemini-1.5-pro	0.74	0.07	550ms	\$0.0055
qwen-vl-7b	0.62	0.10	1500ms	\$0.0000
llava-1.5-13b	0.55	0.12	1900ms	\$0.0000

4.2 3.2 Task generation

For each task (docvqa, chartqa, mmmu) we generate 30 items split across difficulty 1-5 and the task-appropriate question types (docvqa: factual/ocr/reasoning, chartqa: counting/comparison/factual, mmmu: reasoning/factual/comparison).

4.3 3.3 Five chart types

1. **Per-(provider, task) accuracy heatmap**
2. **Accuracy vs difficulty curves** (one line per provider)
3. **Cost vs accuracy frontier** (Pareto scatter)
4. **Per-question-type radar** (one polygon per provider)
5. **Confidence calibration** (reliability diagram)

5 4. Data

Synthetic, deterministic. Real-data drop-in for DocVQA / ChartQA / MMMU is a one-loader change in tasks/load.py.

6 5. Evaluation Setup

5 providers × 3 tasks × 30 items = 450 calls. Hardware: Apple M-series CPU. Mock providers run essentially instantly; real providers would take ~10 minutes of API time at this scale.

7 6. Results

model	n	accuracy	total cost
mock/claude-3.5-sonnet	90	0.633	\$0.7650
mock/gpt-4o	90	0.644	\$1.0800
mock/gemini-1.5-pro	90	0.522	\$0.4950
mock/qwen-vl-7b	90	0.422	\$0.0000
mock/llava-1.5-13b	90	0.278	\$0.0000

model	n	accuracy	total cost
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Three findings:

1. **GPT-4o tops the leaderboard** at 0.644 accuracy. Claude-3.5-Sonnet is essentially tied (0.633) at 71% of the cost.
2. **Cost-efficient pick: Claude-3.5-Sonnet** at \$0.77 for the same workload that costs \$1.08 on GPT-4o. ~30% cost reduction for ~1.7 point accuracy loss.
3. **Open-source mocks (Qwen-VL, LLaVA)** are at \$0 but well behind on accuracy; useful for self-hosted deployments where API spend is non-negotiable.

The reliability-diagram (confidence calibration) shows the expected mild over-confidence pattern across all five providers — they report confidences slightly higher than their observed accuracy across the 0.4-0.8 band.

8 7. Ablations

The mock difficulty penalty was tuned so that the difficulty curves have a meaningful slope (0.05-0.12 per difficulty level). Higher penalties produce noisier curves; lower penalties produce flat ones that defeat the chart's purpose.

9 8. Discussion

The five-chart set together answers the production questions for VLM provider selection. The heatmap tells you which model is best per task. The difficulty curves tell you which model holds up on hard items. The cost-vs-accuracy frontier picks the Pareto-optimal provider. The question-type radar tells you whether one provider is weak on a specific reasoning type. The calibration diagram tells you whether confidence scores are safe to use as a gating signal.

10 9. Limitations

1. **Mock provider profiles are stylized, not measured.** Real numbers will move; chart shapes hold.
2. **Synthetic items.** Real DocVQA / ChartQA / MMMU loaders are not yet integrated.
3. **No batching.** One call per item; real production should batch.
4. **No LLM-as-judge.** Free-form items (DocVQA has many) need a judge for ANLS-style scoring; future work.

11 10. Future Work

- ☐ Real provider adapters (Anthropic, OpenAI, Google SDK calls).
- ☐ HF LLaVA / Qwen-VL providers via transformers pipeline.
- ☐ Real DocVQA / ChartQA / MMMU dataset loaders.
- ☐ LLM-as-judge for free-form answers.
- ☐ Per-difficulty cost-per-correct (the useful efficiency metric).

12 11. References

- Bai, J., et al. (2023). *Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond*. arXiv:2308.12966.
- Beyer, L., et al. (2024). *PaliGemma: A versatile 3B VLM for transfer*. arXiv:2407.07726.
- Liu, H., et al. (2023). *Visual Instruction Tuning*. NeurIPS.
- Masry, A., et al. (2022). *ChartQA: A Benchmark for Question Answering about Charts with Visual and Logical Reasoning*. ACL.
- Mathew, M., et al. (2021). *DocVQA: A Dataset for VQA on Document Images*. WACV.
- Yue, X., et al. (2024). *MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI*. CVPR.

13 Appendix A. Reproducibility

- Repo: Akshitha024/vision-language-model-benchmark, MIT.
- Reproduce: make bench && make plots.
- 5 charts in results/figures/.
- Test artifacts in docs/test_results/.